Battery degradation in municipal e-vehicle fleets for MBTA electrification

Paul Nguyen, Michael Sama, Andrew Wald, Kirsten Moy, Chanwoo Kim
Boston University
ENG ME 306 - Materials Science
Prof. James Chapman
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1 Application Background

The city of Boston is among the top cities in the United States for public transportation [1]. The Massachusetts Bay Transportation Authority (MBTA) provides services for the bus, subway, commuter rail, and ferry. A number of its buses run over 170 routes in the Greater Boston area.

This bus service is in the midst of “one of the most ambitious fleet transition timelines in the nation with plans to electrify its entire fleet of 1,150 buses by 2040” [2]. The implementation of new Battery Electric Busses (BEB) requires not just the replacement of the current hybrid buses. It also requires a complete reconfiguration of bus maintenance facilities and charging systems. The charging protocols of the electric bus’s Lithium Nickel Manganese Cobalt (NMC 111) batteries play a strong part in the battery’s health, ability to store charge, and lifetime.

After the recent series of harsh snowfalls, including the Blizzard of 2026, it is apparent that the MBTA bus fleets need to be prepared for the various temperatures and weather conditions Boston experiences. At the same time, the transportation service needs to keep up with the commuters and resident population of 675,000 who rely on the MBTA’s bus service to get to work, school, and home while also keeping them safe.

The charging protocol must therefore be determined with all of these considerations in mind. If the buses are charged too slowly, they will not be able to keep up with the constant flow of transit riders. If they are charged too fast at high currents, especially in the cold Boston weather, they can create risks like lithium plating. If the buses are too often going from charges close to 0%, all the way up to 100%, and then back down to 0%, the battery’s electrolytes will

decompose faster and its cathodes will degrade quicker. Non-optimal charging practices like these only serve to risk safety and age the batteries. An analysis on the optimal protocol for charging MBTA e-vehicle fleet batteries is extremely important for avoiding battery degradation and for the safety of the population of Boston.

2 Structure-Property Relationships

Lithium Nickel Manganese Cobalt (NMC 111) batteries serve as the primary battery used in municipal fleets of MBTA BEBs. These lithium-ion batteries are composed of a lithium nickel manganese cobalt oxide cathode ($LiNi_{1/3}Mn_{1/3}Co_{1/3}O_2$), a graphite anode, and an electrolyte containing a lithium salt (typically $LiPF_6$) that allows lithium ion transport between electrodes [3]. The performance of this chemistry is controlled primarily by the crystal structures of the cathode and anode materials and how these structures allow for reversible lithium ion intercalation.

The NMC cathode uses a layered α-$NaFeO_2$-type structure consisting of alternating lithium layers and transition-metal oxide layers in which Ni, Mn, and Co occupy octahedral sites within an oxygen lattice [3,4]. This layered ordering creates two-dimensional diffusion pathways that allow Li^+ to move efficiently through the crystal without collapsing the host structure [5]. The oxygen framework maintains lattice strength during cycling, while the transition metals undergo reversible oxidation-state changes to maintain charge neutrality [3]. Each metal contributes a specific structural function. Nickel provides high capacity to the battery due to its accessible valence states that allow for more extraction of Li^+ at relatively low potentials, increasing the overall energy density of the battery. Cobalt enhances the electronic conductivity and improves

Li^+ mobility between layers due to its favorable electronic structure. Manganese provides structural stability by suppressing lattice distortion. The equal 1:1:1 ratio of Ni:Mn:Co therefore balances the three main properties of capacity, safety, and cycle life which is critical for high-duty-cycle transit applications like the MBTA BEBs [3]. The graphite anode also has a layered structure, composed of stacked graphene sheets held together by van der Waals forces in a hexagonal lattice [5]. During charging, lithium ions leave the NMC cathode layers, migrate through the electrolyte, and intercalate between the graphene planes in the graphite anode. Electrons simultaneously travel through the external circuit to maintain charge balance [3,5]. During discharge, Li^+ de-intercalate from the graphite and reintegrate into the cathode's lithium layers, allowing for electrons to flow through the motor circuit [5]. Because both electrodes rely on layered intercalation hosts, the battery can undergo many cycles with minimal structural collapse, given that the lattice stays intact.

These structural features directly determine the macroscopic properties of NMC 111 batteries. Layered crystal ordering enables fast Li^+ diffusion and high rate capability, which supports rapid acceleration and regenerative braking demands in electric buses [3,4]. The metal-oxide octahedral framework and stable oxygen lattice provide mechanical chemical stability, supporting long cycle life [5]. Interlayer spacing in both cathode and anode controls lithium mobility and therefore influences power density and internal resistance [3]. Under ideal operating conditions, the layered host structures accommodate many different strains reversibly, such as local volume changes within secondary particles or fluctuations in transition-metal redox. However, in real-world transit service, particularly under the cold winter climate and

high-charging-rate demands of Boston bus operation, these pathways become the primary drivers of degradation.

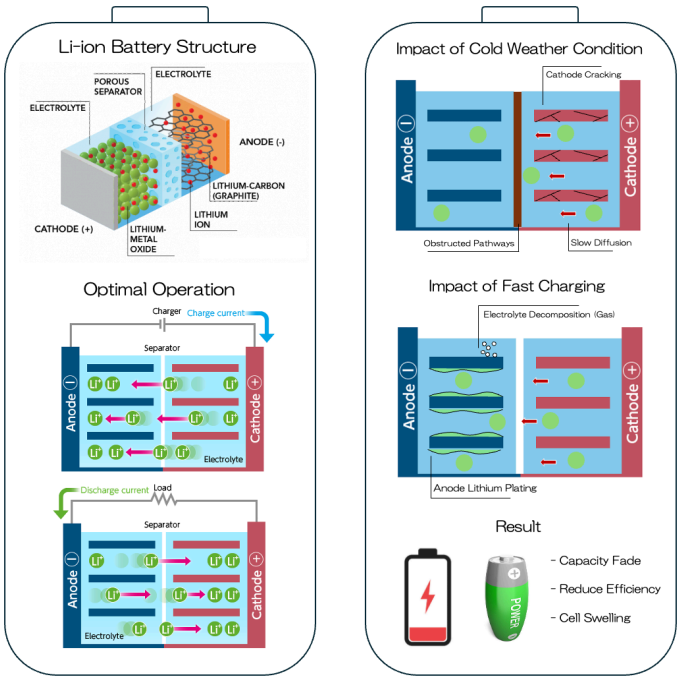


Figure 1: Structure-Property Relationships and Degradation Mechanisms of NMC 111. (Left) The layered oxide structure provides 2D pathways for efficient lithium-ion diffusion. (Right) Operational stress from freezing temperatures and high C-rate charging (3C–4C) triggers coupled degradation pathways.

The cold weather climate of the city of Boston and the charging speeds necessary for public transportation applications place tremendously challenging operational conditions on the batteries at hand. The coupled mechanical and electromechanical degradation pathway that forms as a result of these conditions accelerates structural failure and shortens battery life. The cold weather operational conditions present in Boston significantly alters the battery performance, both the short term and long-term cycling. At reduced temperatures, lithium diffusion within

both the cathode and anode slows substantially, increasing internal resistance and charge-transfer polarization during charging [6]. This leads to greater mechanical stress within secondary particles as a result of inhomogeneous lithium insertion and removal. The elevated stress concentrations at grain boundaries and pre-existing internal defects accelerate crack initiation and propagation interfacial reactions and accelerating electrolyte consumption [6]. Isolation of active material due to fracturing in tandem with lithium plating promotes the formation of electrically inactive “dead lithium,” reducing the inventory of cyclable lithium within the battery cell [7]. Over time enduring these conditions, deposits can accumulate, obstructing porous pathways and separator interfaces, and impeding ion transport while further increasing cell impedance. Overall, these conditions upon repeated cycles accelerates capacity loss, increases resistance, and creates structural instability to an elevated magnitude as compared to normal operational temperatures.

Fast charging has similar compounding effects on capacity loss and instability, with the dominant aging mechanism depending on the applied charging rate of the battery. Charging rates are expressed in units of “C”, where increasing C-rate corresponds to reduced charge time and higher current density. Experimental studies demonstrate that degradation pathways have changing dominant contributors as charging rate increases [8,9]. At moderate charging rates of 1-2C (30-60 minute charge times), degradation is governed by mechanical particle cracking in the cathode [9]. Rapid lithium insertion generates internal concentration gradients and anisotropic lattice strain, producing stress within secondary particles. Over extended cycling, the accumulated stress promotes crack initiation and propagation as previously described [9]. Although degradation occurs here, it is less severe than the pathways seen at higher C-rates due

to the absence of substantial lithium plating. Nevertheless, charging at these rates is insufficient for the application of rapid turnover public transportation electrification efforts which cannot afford to wait. At an increased charging rate of around 3C (20 minute charge time), degradation is dominated by lithium plating on the anode [8,9]. Under this high current density, lithium ions cannot intercalate into the graphite of the anode quickly enough, leading to lithium deposition on the anode surface. As a result, cycle life is reduced, cyclable lithium is depleted, and impedance is increased [8]. At an extreme fast charging rate of around 4C, electrolyte decomposition becomes increasingly pronounced, and is accompanied by gas evolution. In particular, by the formation of ethylene gas at the graphite electrode [9]. Gas generation in a battery cell increases internal pressure, degrades the electrode-electrolyte interface, and further contributes to greater capacity fade and degradation.

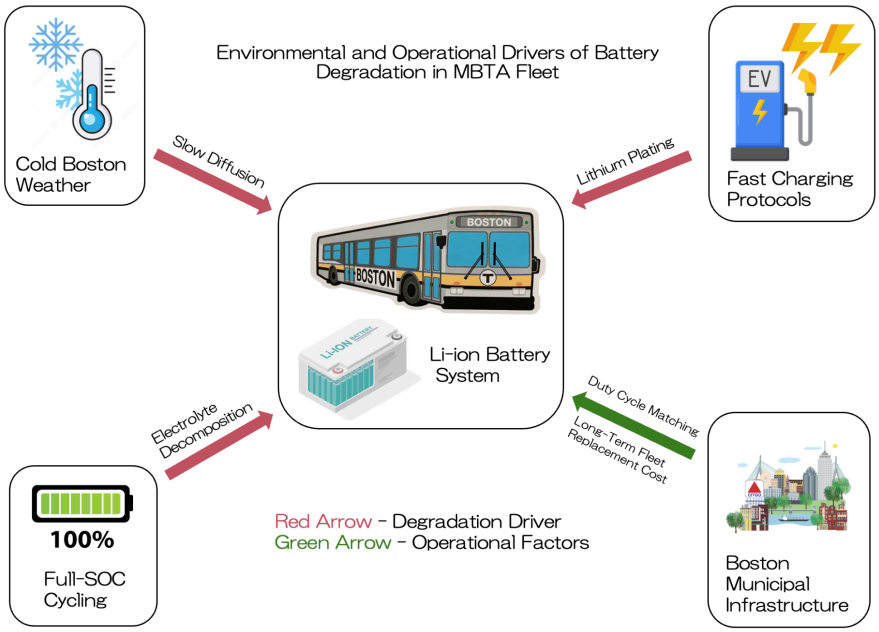


Figure 2: Environmental and Operational Drivers of Battery Degradation in MBTA Fleet. This infographic illustrates the external stressors that drive internal material degradation mechanisms

3 Synthesis and Processing

Lithium Nickel Manganese Cobalt (NMC 111) is synthesized through a multi-step process that involves co-precipitation, calcination, and electrode manufacturing. Each of the materials in the battery have different properties that influence its behavior and performance. The nickel has a high specific energy, so it greatly contributes to the battery's energy density[17] . However, it can be unstable and needs to be balanced out by other elements, such as manganese and cobalt. Manganese helps provide the composition with structural stability and lower its internal resistance[12]. This helps with the battery's lifespan and cycle capabilities, making it safer for long term use. Cobalt helps stabilize the composition and increases its specific energy, helping with energy storage and adds to the battery's density [14].

Hydroxide co-precipitation is a cost-effective method used to ensure the mixture is combined homogeneously [14]. Equal parts of nickel, manganese, and cobalt balance the energy efficiency of the lithium batteries which is why they are synthesized in a 1:1:1 ratio [13]. These metal ions precipitate simultaneously to form a precursor material for the lithium battery. The process starts by dissolving transition metals (NMC sulfates) in water and adding a precipitating agent, which can be either sodium hydroxide or ammonium hydroxide. Aside from the ion concentrations, temperature, pH, and stirring rate are also important conditions for the synthesis of the material. Specialized continuous stirred-tank reactors (CSTR) are used to closely monitor the metals as they precipitate. At pH levels below 10, Ni precipitates slower than Mn and Co, and at pH levels above 12, impurities can be trapped, so the pH level is monitored precisely during this phase. High temperatures may also lead to impurities and irregularities and the stirring speed impacts composition as particles will settle unevenly if it's not regulated properly.

Since the metals precipitate at the same time, they form evenly distributed particles. Each particle is homogeneous and allows for better battery performance with more cycles and storage capabilities [10]. $Ni_xMn_yCo_{1-x-y}(OH)_2$ is the resulting precursor. Co-precipitation is a commonly used process that's reliable and allows for greater particle control [14]. Additionally, the resulting material property is consistent in large batches due to the amount of parameters controlled during the process. After precipitation, Lithium is mixed into the composition, then the solution is heat treated, through a process called calcination, to form the crystalline ($LiNi_xMn_yCo_zO2$). Calcination is a process of removing impurities by raising the compound to high temperatures without melting it [15]. The oxygen supply to the compound is restricted, which helps regulate thermal decomposition and oxidation. If too much oxygen exists in the compound, it can lead to weaker crystal structures, cracks, and a shorter battery lifespan.

The battery is assembled through electrode manufacturing [14]. The NMC mixture is spread onto metal foils then sized to fit in a battery. Cathodes are spread onto aluminum foil and anodes are spread onto copper foil to properly collect current. Electrodes are stacked with separators to prevent shorting the battery. The NMC mixture is drawn into the holes of the electrodes using vacuums and finally, the battery is welded shut. Lithium ions move from cathode to anode through the electrolyte and flow back when the battery is in use [11]. Batteries are used many times so their structural composition needs to account for battery usage. If the NMC layers are too thick, lithium ions would not be able to travel through them, preventing the battery from charging and discharging properly. Finally, the battery is stored at a constant temperature and checked regularly to ensure proper usage.

4 Operation Conditions

The environment for which public transportation in the city of Boston is forced to operate within is inherently challenging, with large temperature swings as a result of its hot summers and cold winters, disorganized roadway logistics, high traffic demands, and inconsistent infrastructure. As a result, the existing transportation systems are already strained often beyond their limits, resulting in a reputation for unreliability that the MBTA constantly battles. However, with growing global environmental and energy concerns leading to a shift towards renewable energy to power public transportation in the city, a new set of challenges centered around battery technologies has been introduced as the MBTA adds electric buses to its fleet.

Many of the challenging environmental factors that affect Boston uniquely hamper the NMC 111 battery's performance that have been introduced in the electric buses recently added to the MBTA fleet. Operating conditions play a critical role in determining how battery materials evolve structurally over time, and these structural changes directly govern performance. At low temperatures characteristic of Boston's winters, lithium-ion diffusion within the electrode materials slows exponentially [16]. This reduced diffusivity limits how quickly lithium ions can intercalate into the electrode's ordered crystalline lattice, slowing charge transfer and increasing internal resistance, exacerbated over the rapidly repeated charge-discharge cycles required of continuously operating public transportation vehicles [17]. Additionally, low temperatures intensify mechanical stresses at existing grain boundaries and pre-existing defects within the electrode microstructure. These stress concentrations promote cracking, disrupting the continuous pathways needed for lithium transport [33].

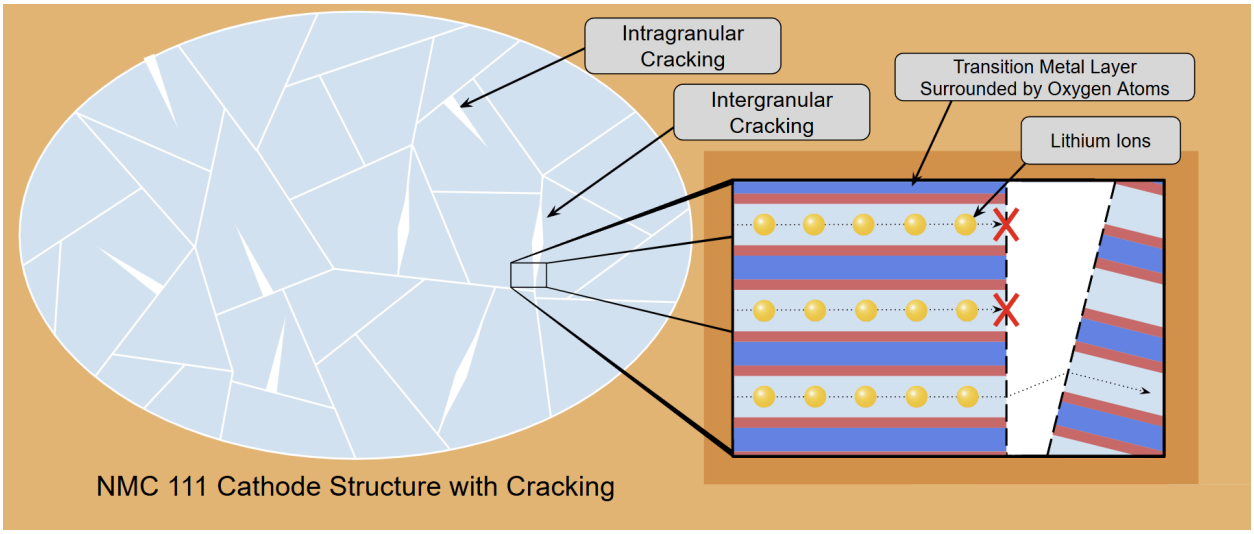


Figure 1: Intragranular and Intergranular Cracking in NMC111 Cathode Crystal Structure. The image shows the grains and grain boundaries with intergranular (between grains) and intergranular (within grains) cracking. To the right, it zooms in to an intergranular crack, showing how the crack can disrupt lithium ion transport pathways.

These material structural disruptions also contribute to lithium consumption mechanisms. Instead of cleanly intercalating into the crystalline structure, lithium can deposit as “dead” lithium on the anode, forming inactive metallic regions that can no longer participate in electrochemical cycling. This accumulation of lithium is often associated with defects in the electrolyte and electrolyte interfaces, where uneven flux promotes localized disposition. Structurally speaking, the initially ordered crystalline framework of the electrode is essential for efficient lithium diffusability. Once this structure is compromised by cracking or deposition, ionic transport pathways become blocked or disrupted, leading to reduced capacity and efficiency.

The situation for which buses must operate under as constantly operating public transportation devices also play a large role in the material performance of its batteries from a structural basis.

Electric buses at current technological levels have comparably low operational range as compared to their gas counterparts, and thus must charge more frequently to have similar usability. As charging is a delayed process in the continuous operation cycle of the bus as a part of the public transportation system, minimizing the charge time would benefit its usability. However, increasing the charge rate of NMC 111 batteries has varying and increasingly degrading implications without considering previously described temperature effects. At moderate charge rates (1-2C), rapid lithium insertion and extraction generates internal concentration gradients and anisotropic lattice strain, producing stress within secondary particles and subsequent cracking [34]. More specifically regarding the anisotropic strain, the lattice structure expands and contracts differently along different crystallographic axes during lithiation and delithiation, and the mismatch in axes creates forces that promote intergranular cracking. This increases the surface area exposed to the electrolyte, accelerating side reactions. At increased charge rates (~3C), lithium ions cannot intercalate quickly enough into the anode, resulting in the accumulation of inactive lithium deposits on the surface. When metallic dendrites “strip” during discharge, they often fracture at the base, leaving isolated and electronically disconnected metallic lithium that is no longer cyclable [20].

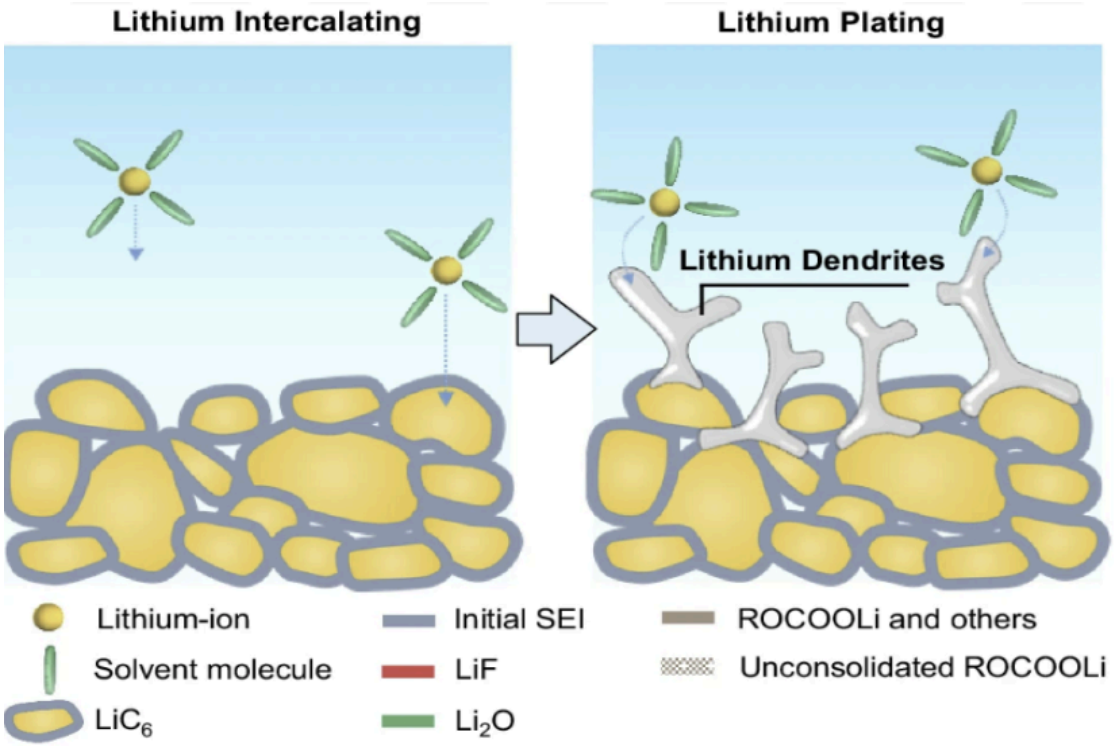


Figure 2: Formation of Lithium Dendrites Under Fast Charging Rates (~3C). At fast charging rates, lithium ions are not able to intercalate fast enough, causing lithium plating. Deposits of lithium accumulate at the surface. This can cause isolated islands of inactive lithium, which can no longer cycle [20].

At more extreme charging rates (~4C), oxygen evolution from the cathode can occur. At high delithiation rates, the bonds between the transition metal and oxygen become weak, allowing oxygen to be released from the lattice, triggering the collapse of the layered structure. The release of oxygen drives a phase transformation from the layered rhombohedral structure to a spinel structure, then to a rock salt phase at the particle surface [21]. This rock salt phase is electrochemically inactive, and has a significantly lower electronic and ionic conductivity, forming a resistive outer layer that impedes transport. Transit bus operation subjects onboard NMC 111 battery packs to continuous mechanical vibration arising from road surfaces

irregularities, drivetrain dynamics, and urban stop-and-go cycles. Unlike degradation driven by electrochemical cycling, mechanical vibration does not introduce fundamentally new degradation mechanisms; rather, it accelerates the onset and progression of structural damage that would otherwise develop over thousands of charge-dischare cycles [22, 23]. The consequences of this acceleration are felt across multiple length scales, from the grain boundaries within individual NMC secondary particles to the macroscale electrode-current collector interface, and they translate directly into measurable losses in capacity and power capability.

On the particle level, the crystalline secondary particles of the NMC 111 cathode are vulnerable to vibrational stress because the layered structure expands and contracts anisotropically during cycling. When vibration loads are superimposed onto this existing lattice strain, stress concentrates at grain boundaries between primary particles, driving intergranular cracking and eventual particle fragmentation [23, 24]. The graphite anode experiences similar damage. Specifically, vibrations at 50Hz have been shown to induce visible cracking and surface changes on graphite electrodes, with sediment accumulation observed on the separator following subsequent cycling, which indicates physical shedding of graphite fragments [23].

Particle fragmentation exposes fresh NMC and graphite surfaces to the electrolyte, substantially increasing the reactive surface area. This drives accelerated transition metal dissolution and continuous electrolyte decomposition at newly formed interfaces, replacing the stable passivating SEI with a thicker, more resistive surface film [23, 24]. This consumes cyclable lithium and increases charge transfer resistance at the electrode-electrolyte interface, directly linking grain-boundary-scale mechanical damage to capacity fade and impedance rise at the cell level

[24]. At the electrode assembly scale, repeated stress cycles degrade adhesion between the active material coating and current collector, causing electrode delamination and loss of electrical contact [23]. This isolates regions of NMC or graphite from the conductive network, constituting an effective loss of active material even without chemical degradation of the material itself. The combined result of increased surface film resistance, contact loss, and active material isolation is higher internal resistance, reduced power capability, and accelerated capacity fade. These mechanisms compounded over the service lifetime of MBTA fleet buses by continuous daily vibrational loading [22, 23, 24].

5 Performance Changes

5.1 Temperature

Low-temperature operation significantly degrades battery performance compared to its initial state at room temperature. These effects are primarily driven by reduced lithium-ion diffusivity, increased internal resistance, and structural damage within electrode materials [25, 26].

One of the most critical changes is the reduction in lithium diffusivity, which limits ion transport in both the anode and cathode. As temperature decreases, ionic mobility drops [26, 27], slowing intergalactic kinetics and increasing polarization and voltage losses, especially at high rates [27]. Experimentally, batteries operating around −20 °C can experience a ~39% reduction in capacity compared to room temperature [25], as lithium ions cannot fully access active material during cycling. Internal resistance also increases due to reduced ionic conductivity and the accumulation of degradation products [27]. Lithium plating and electrolyte decomposition form insulating layers on electrode surfaces, impeding change transport [25]. This results in larger voltage drops,

reduced power output, and lower energy efficiency [27], with resistance continuing to grow under repeated low-temperature cycling [25]. Structural degradation further worsens performance. Cracking a train's boundaries leads to loss of active material (LAM) [28], while lithium plating forms electrically inactive dead lithium, reducing lithium inventory (LLI) and causing irreversible capacity fade [25]. These effects decrease the amount of lithium available for reversible reactions. Additionally, accumulated deposits obstruct porous pathways within the electrode and separator, further increasing impedance and reducing rate capability [25]. Over repeated cycles, these combined effects accelerate degradation and significantly shorten battery lwife [25, 28]. Compared to its initial state, the battery ultimately exhibits lower capacity, higher resistance, reduced power capability, and decreased durability under low-temperature conditions.

5.2 Electrolyte

Degradation in lithium ion batteries is primarily caused by the growth of the solid electrolyte interphase (SEI) layer [29]. During charging and discharging cycles, SEI growth consumes cyclable lithium ions, which leads to capacity degradation [29]. Additionally, oxide solid electrolytes can form grain boundary resistance, which reduces the battery's power density [30]. SEI growth is represented by the reaction:

$$S + ke^{-} + kLi^{+} -> P$$

where S is the solvent, P is the formed insoluble product, and k is the number of electrons transferred per reaction [31]. The resulting capacity loss is represented in the following equation:

$$x(t) = \frac{N_{lost}}{N_0} = 1 - \frac{C_{Li}(t)}{C_0}$$

Where N_0 is the initial mole number of cycling lithium, $N_{Li}(t)$ is the available mole number of lithium ions at time t, and C_{Li} is the molar concentration of lithium [31]. Lithium loss due to SEI growth is expressed as:

$$N_{Lost} = Z_P \cdot C_P \cdot A \cdot (L(t) - s)$$

Where Z_P is the stoichiometric coefficient of lithium in P, C_P is the density of P in the SEI, A is the surface area of the anode, L(t) is the thickness of the SEI layer at time t, and L(0) = s is the initial thickness [31].

From these equations, electrolyte degradation from SEI layer growth reduces lithium available for charging cycles, lowering long term battery efficiency [29]. Oxide electrolyte grain boundaries hinders lithium ion movement by forming areas with high resistance [30]. Both SEI growth and grain boundary effects reduce lithium ion battery efficiency, capacity, and long term usage.

5.3 Fast Charging

C rate refers to the rate that a battery is charged and discharged at, with fast charging occurring at high C rates [32]. While fast charging increases charging speed, it also increases stress and heat generation, which reduces overall battery performance. Higher C-rates have high peak temperatures and current densities, which increases polarization at the electrodes [32]. Polarization is the voltage difference that drives electrochemical reactions beyond the required voltage, causing capacity loss and more heat generation. Fast charging also leads to lower battery capacities. For example, a 4C battery experiences a retention drop to 67.3% while a 2C charge resulted in only a 2.1% capacity drop [32]. The degradation from fast charging compromises the

battery's performance by causing cathode particles to crack and promote SEI growth. Additionally, fast charging can lead to lithium plating on the anode surface, increasing reactions between the lithium and electrolyte layers. This produces heat and gas, which prevents battery heat dissipation and leads to fires [32]. Lithium plating is more likely to occur at low temperatures, making fast charging dangerous in cold areas such as Boston [32].

5.4 Vibration

Mechanical vibration significantly accelerates battery degradation by introducing additional mechanical stress to electrode materials, even though it does not create entirely new degradation mechanisms [36]. Instead, vibration amplifies existing failure pathways such as particle cracking, lithium loss, and impedance growth, leading to faster performance deterioration compared to normal operating conditions.

One major structural effect of vibration is electrode particle cracking and fragmentation. Cyclic mechanical loading from vibration, around 50Hz, induces microcracks and fractures in both cathodes and anode materials [34]. These cracks increase reactive surface area and promote side reactions. Studies have shown that vibration exposure can result in an additional 5-10% capacity loss compared to non-vibrated cells under the same cycling conditions [34]. These structural changes directly alter material properties. Cracking and particle separation reduce electrical connectivity, limiting electron transport pathways Vibration has also been shown to increase internal resistance by 10-20% due to contact loss and interface degradation [33]. Additionally, vibration accelerates loss of lithium inventory (LLI) and loss of active material (LAM), both of which contribute to long-term degradation [35]. As a result, these property changes lead to clear

performance losses. Increased resistance causes larger voltage drops and reduced power capability. While lithium loss reduces usable capacity. Experimental studies report that cells subjected to vibration exhibit higher impedance growth rates and faster capacity fade per cycle, especially under combined cycling conditions [33, 34]. Over repeated cycles, vibration-induced damage accumulates and significantly shortens battery life. Compared to its initial state, the battery shows lower capacity, higher impedance, and reduced efficiency. Overall, mechanical vibration does not introduce new degradation mechanisms but accelerates existing ones, resulting in faster performance loss and reduced durability in real-world applications [36].

6 Performance Improvement

6.1 Mitigation of Intergranular Cathode Cracking

A main degradation pathway in NMC 111 cathodes under MBTA operating conditions originates from the polycrystalline structure of secondary particles, where each particle consists of many smaller grains separated by grain boundaries. These interfaces act as mechanically weak regions due to crystallographic misorientation and anisotropic lattice expansion during lithiation and delithiation. Under repeated cycling and continuous mechanical vibration, stress localizes at these grain boundaries, promoting intergranular crack initiation and propagation. As cracks form, they disrupt lithium-ion transport pathways, increase exposure of fresh surfaces to the electrolyte, and accelerate reactions like electrolyte decomposition and transition metal dissolution. This leads to a loss of active material and increased internal resistance.

Transforming the cathode microstructure so that each particle behaves as a single continuous crystal directly addresses these issues by eliminating internal grain boundaries. In doing so, this

directly addresses these failure mechanisms at their structural origin. By eliminating internal grain boundaries, the dominant mode of mechanical failure shifts from intergranular fracture to intragranular. As intergranular cracking requires substantially higher stress to initiate and propagate, there will be no pre-existing weak interfaces to guide crack growth. As a result, stress induced by anisotropic lattice expansion during cycling is distributed more uniformly throughout the particle volume, rather than concentrating at discrete boundary regions. This leads to measurable improvement in mechanical integrity under both electrochemical and vibrational loading, reducing the rate of particle fragmentation and preserving structural continuity over extended cycling. In addition, the absence of grain boundaries limits the formation of new reactive surfaces, suppressing electrolyte decomposition and the growth of resistive surface films, thereby reducing impedance rise over time. This structural coherence also enhances stability under high C-rate (charge/discharge rate) operation. In polycrystalline materials, grain boundaries act as initiation sites for phase transformations from the layered structure to spinel or rock-salt phases, which are electrochemically inactive and impede in lithium transport. By reducing these defect sites, a single crystal architecture helps preserve the layered structure, maintaining more consistent ionic and electronic conductivity. This is particularly important under conditions of rapid charging and discharging, where internal strain gradients are more severe and degradation mechanisms are accelerated.

However, these improvements come with trade-offs. Larger, single-crystal particles increase lithium diffusion path length and reduce surface area, which can limit lithium transport kinetics and reduce rate capability during fast charge and discharge cycles. Additionally, non-uniform particle sizes can introduce uneven lithiation and internal stress gradients, potentially

reintroducing cracking through different mechanisms. Structural disorder, such as transition metal migration into lithium layers, can further hinder lithium diffusion and reduce overall efficiency. As a result, while this approach significantly improves resistance to vibration-induced damage and cracking, it may reduce performance under high-power operating conditions, requiring a balance between mechanical durability and electrochemical performance.

6.2 Cathode Doping for Fast-Charging and Thermal Stability

Lithium Ion batteries are negatively impacted by temperature, vibrations, and fast charging. The degradation from fast charging compromises the battery's performance by introducing heat and stress to the cathode layer, causing it to crack. Varying environmental temperatures also affect the diffusion rate of ions moving across the cathodes. Adding impurities to the NMC layers will improve battery performance through increased resistance to fast charging and temperature effects. Introducing small amounts of alloying elements to the NMC mixture will distort the solution's crystal lattice structure and reduce dislocation motion. Although this may reduce the material's ductility, it can increase its strength and hardness, allowing it to withstand vibrations and fast charging for longer periods of time. Depending on which elements are introduced to the material, its grain size will change, affecting its conductivity and ion diffusion rates. Fine precipitates form from impurities which block dislocation motion, increasing diffusion rates by opening up more pathways for ions to travel through. This also reduces permanent deformation in the cathode layer and therefore, the battery can maintain high performance over longer periods of time.

Doping cathode material helps improve fast charging performance by increasing structural stability under high current and heat generation. If the crystal structure remains more intact during fast charging, the electrode experiences less cracking and less permanent damage, which can help maintain a more stable lithium pathway and reduce degradation.

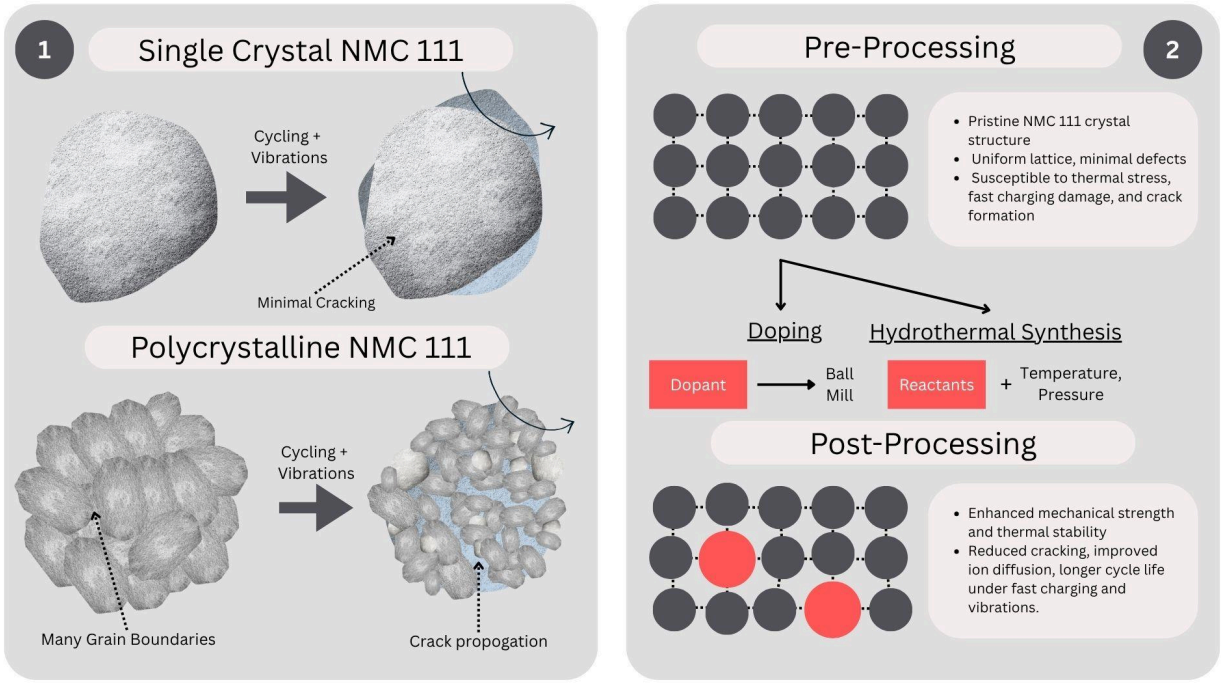


Figure 1: Description of both solutions' structure-property relationships and added benefits. Solution 1 (left) shows the difference in cracking present after repeated cycling and vibrations between single-crystal and polycrystalline NMC material. Solution 2 (right) shows the impact of doping and hydrothermal synthesis on NMC material.

7 Processing & Synthesis

7.1 Nucleation and Calcination for Mechanical Vibration

As discussed in the first proposed solution, promoting larger grains and reducing grain boundary area leads to a structurally stronger cathode material with improved resistance to cracking and

degradation. This transition from a polycrystalline to a single-crystal like structure NMC cathode structure is achieved by controlling synthesis conditions to favor particle growth over nucleation during the formation stage. By lowering the nucleation rate in the co-precipitation process, fewer particles are initially formed, allowing them to grow larger and denser over time. This reduces the likelihood of forming agglomerates composed of many small primary grains and instead promotes particles with greater structural continuity. As a result, the internal grain boundary density within each particle is significantly reduced, moving the material closer to single-crystal behavior. Additional control through hydrothermal synthesis provides a regulated environment for temperature and reaction time, helping to suppress uncontrolled nucleation and ensure more uniform particle growth and compositional consistency.

Following particle formation, calcination is used to further refine the crystal structure by promoting grain growth and coalescence. Higher temperatures and longer dwell times provide the thermal energy needed for smaller crystallites to merge into larger, continuous domains, reducing crystallographic misorientation and eliminating internal interfaces that act as crack initiation sites. This step enhances overall crystallinity and structural integrity while preserving the layered NMC framework required for lithium-ion transport. Together, these synthesis modifications would produce larger, more uniform particles with fewer internal defects, resulting in a cathode material that is mechanically more robust and less susceptible to intergranular degradation, while keeping the majority of its electrochemical functionality.

7.2 Dopant Incorporation through Ball Milling and Hydrothermal Synthesis

The addition of select chemical species with the desired properties can be achieved with doping or mixing during the hydrothermal synthesis process.

For batteries made through a sintering process, a doping step could be inserted beforehand. The base ceramic composition powder of the battery cell is mixed with small amounts of dopant powder, composed of chemical species with properties tending to higher electrical conductivity and mechanical strength. Ball milling is essential to mix in the dopants more uniformly and break up any agglomerates. The doped powder is then densified through granulation, pressing, and sintering while preserving the chemical homogeneity achieved through mixing. This results in the NMC 111 ceramic structure with additional chemical species that may increase the electrical conductivity and mechanical strength.

Another option for synthesis is during the hydrothermal synthesis of the ceramic battery structure. In hydrothermal synthesis, the reactants are dissolved, mixed, and placed in a hydrothermal reactor. Under high temperature and pressure conditions, they form into a new ceramic structure. The desired chemical species can be added into the initial solution of reactants, allowing the ceramic structure to form with those species. This again results in the NMC 111 ceramic structure with certain additional chemical species, chosen to increase electrical conductivity and mechanical strength.

8 Solution Judgement

Taking a look at both solutions, they both address key stress factors on the performance of the batteries. Solution 1, lowering the nucleation rate, reduces the grain boundary area which results

in a ceramic structure more resistant to cracks from mechanical vibration. Solution 2, adding select chemical species either in the solution for hydrothermal synthesis or through doping, can allow the NMC 111 structure to achieve higher electrical conductivity and mechanical strength, improving resistance to fast charging processes and temperature driven degradation.

Both solutions offer meaningful improvements, but each has their trade-offs. Solution 1 could lower the amount of diffusion pathways for lithium ions and lower the charging rate capabilities, while solution 2 relies on choosing the correct chemical species to add and in what concentration. This would make solution 2 highly reliant on great amounts of testing and additional research. It is because of this that we think solution 1 is the most feasible. It offers a solution that addresses what is most likely the primary and most consistent stressor in continuous road use, mechanical vibrations. It also has many fewer variables to account for than solution 2. The viability of solution 2 relies on finding the right dopant to achieve the desired qualities through testing, while also ensuring the new structure does not impede the current functionality of the battery. For these reasons, promoting the formation of larger grains to address cracking from mechanical vibrations is the best solution.

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